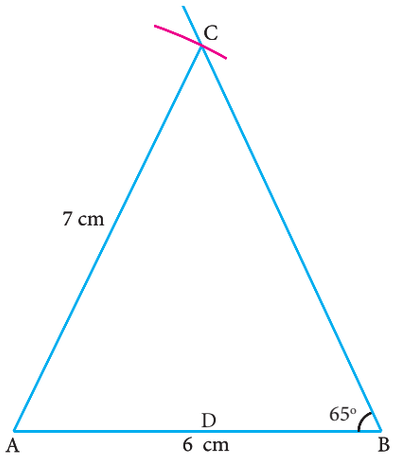


Fig. 4.95

Step 1 : Draw the ΔABC with AB = 6cm, $\angle B = 65°$ and AC = 7cm

Step 2 : Construct the angle bisectors of any two angles (A and B) and let them meet at I. Then I is the incentre of ΔABC. Draw perpendicular from I to any one of the side (AB) to meet AB at D.

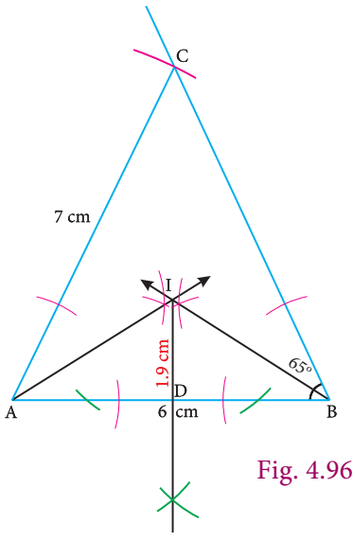


Fig. 4.96

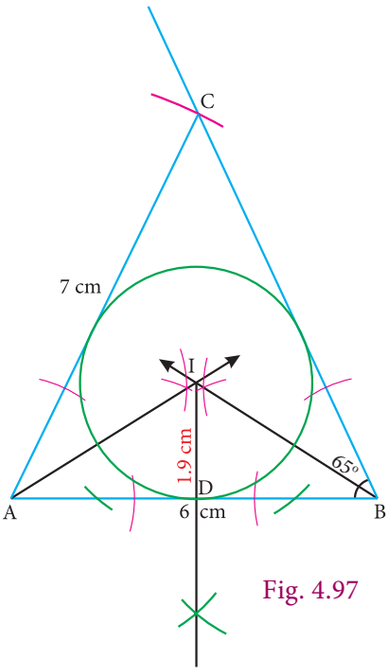


Fig. 4.97

Step 3: With I as centre and ID as radius draw the circle. This circle touches all the sides of the triangle internally.

Step 4: Measure inradius
In radius = 1.9 cm.

Exercise 4.6

1 Draw a triangle ABC, where AB = 8 cm, BC = 6 cm and $\angle B = 70^0$ and locate its circumcentre and draw the circumcircle.

2 Construct the right triangle PQR whose perpendicular sides are 4.5 cm and 6 cm. Also locate its circumcentre and draw the circumcircle.

3. Construct ΔABC with $AB = 5$ cm $\angle B = 100^0$ and $BC = 6$ cm. Also locate its circumcentre draw circumcircle.

4. Construct an isosceles triangle PQR where $PQ = PR$ and $\angle Q = 50^0$, QR = 7cm. Also draw its circumcircle.

5. Draw an equilateral triangle of side 6.5 cm and locate its incentre. Also draw the incircle.

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